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ABSTRACT

The rapid growth of digital communication has led to emojis becoming an integral part of expressing emotions, tone, and context in text-based conversations. However, the manual use of emojis is often subjective and context-dependent, which creates an opportunity for automation through intelligent systems. This project, *Emoji Predictor*, focuses on designing and implementing a machine learning–based model that predicts suitable emojis for short text inputs. The system aims to enhance user interaction and emotional expressiveness by automatically suggesting emojis that best represent the sentiment and intent behind a message.

The project involves multiple stages of data processing and model development. A dataset containing text and corresponding emojis was used for training and evaluation. The text data was preprocessed by cleaning, tokenizing, and encoding before being passed to machine learning models. Two different approaches were explored — a sequential Long Short-Term Memory (LSTM) network and a Bidirectional LSTM model — to identify the most effective method for emoji prediction. Comparative analysis between these architectures provides insights into trade-offs between performance accuracy and computational efficiency.

The LSTM-based approach demonstrated efficient learning of sequential patterns in short texts and provided faster inference times, making it lightweight and suitable for real-time applications In contrast, the Bidirectional LSTM (BiLSTM) model, which processes text in both forward and backward directions, captured a richer understanding of contextual relationships within the input sequences. This dual-direction processing allowed the model to better grasp the nuanced emotional cues in the language, enhancing its ability to make accurate predictions. However, while BiLSTMs can achieve strong performance, they still require considerable computational resources and longer training times compared to simpler sequence models.

To make the system accessible and interactive, a Streamlit-based web application was developed. Users can input any text, and the app predicts the most relevant emoji using the trained model. The interface is simple, responsive, and suitable for real-world demonstrations. This deployment bridges the gap between research and user experience, illustrating how AI-driven natural language understanding can enhance human–computer interaction.

CHAPTER-1

INTRODUCTION

In today's digital era, text-based communication has become the most common form of interaction across social media, messaging platforms, and online communities. However, plain text often lacks emotional depth and context, leading to misunderstandings or neutral interpretations of tone. Emojis play a vital role in overcoming this limitation by visually representing emotions, reactions, and sentiments, thereby enhancing the expressiveness and relatability of digital communication. The use of emojis has evolved from simple facial expressions to a rich symbolic language that complements textual content in everyday interactions.

Despite their popularity, choosing the most suitable emoji to match a message's context remains a subjective task. Users often rely on intuition, and the selection can vary widely based on individual interpretation, culture, and emotion. This challenge opens the door to leveraging Artificial Intelligence (AI) and Natural Language Processing (NLP) techniques to automate emoji prediction. By understanding the semantics and sentiment of a given text, AI models can intelligently suggest or generate the most contextually relevant emoji, thereby improving user experience and communication efficiency.

The *Emoji Predictor* project was developed with this motivation in mind. It aims to design and implement a system capable of predicting appropriate emojis for short text inputs using machine learning models. The project explores two major approaches: a sequential deep learning model using Long Short-Term Memory (LSTM) networks, and a Bidirectional LSTM. The LSTM model focuses on learning sequential dependencies between words, while the Bidirectional LSTM model processes sequences in both directions, capturing richer contextual dependencies to understand deeper linguistic nuances.

CHAPTER-2

METHOD

The methodology adopted for the *Emoji Predictor* project follows a structured approach involving data collection, preprocessing, model development, evaluation, and deployment. Each stage was designed to ensure that the system effectively learns textual patterns and accurately predicts relevant emojis. The entire workflow is illustrated through the following key phases.

1. Data Collection and Exploration

The first step involved collecting and understanding the dataset used for training and evaluation. The dataset contained short text messages (such as tweets, phrases, and sentences) paired with corresponding emojis that represent the intended emotion or context. Exploratory Data Analysis (EDA) was conducted to study the frequency of different emojis, identify class imbalances, and examine linguistic patterns in the text data. Visualization techniques such as bar graphs and word clouds were used to gain insights into emoji distribution and text sentiment trends. This phase ensured that the dataset was suitable for supervised learning and guided later preprocessing decisions.

2. Data Preprocessing

Before training the models, several preprocessing techniques were applied to clean and standardize the data. The text data was converted to lowercase, and unnecessary characters such as punctuation, URLs, special symbols, and user mentions were removed. Tokenization was then performed to split the text into individual words or tokens. For the LSTM model, the Keras Tokenizer was used to create a vocabulary and convert text into integer sequences of a fixed length.. The emojis were label-encoded into numerical categories using the LabelEncoder function. This step ensured consistent input representation across both models.

3. Model Development

Two different architectures were implemented and compared for emoji prediction:

- LSTM-basedModel:

  The Long Short-Term Memory (LSTM) network was chosen due to its capability to capture long-term dependencies in sequential data. The model architecture consisted of an embedding layer to represent words in vector form, followed by LSTM layers to learn contextual relationships. Dropout layers were used for regularization, and a dense softmax output layer produced the probability distribution across emoji classes. The model was trained using categorical cross-entropy loss and the Adam optimizer.

- **BidirectionalLSTMModel**:

  The second approach employed a Bidirectional LSTM, which processes text sequences in both forward and backward directions to capture contextual meaning from both past and future word dependencies. The pre-trained Bidirectional LSTM model was fine-tuned on the emoji dataset by adding a classification head—a dense layer with softmax activation—to predict emoji categories. The model was trained with a smaller learning rate and fewer epochs to prevent overfitting.

4. Model Evaluation

Both models were evaluated using standard classification metrics such as accuracy, precision, recall, and F1-score. Confusion matrices were plotted to visualize misclassifications and identify which emojis were frequently confused. Additionally, inference latency and resource consumption were analyzed to compare model efficiency. The LSTM model provided faster predictions with moderate accuracy.

5. Deployment using Streamlit

To make the system interactive and accessible, a Streamlit-based web application was developed. The trained model was integrated into the web app, allowing users to input text and receive the most relevant emoji prediction in real time. The application features a simple, responsive interface and was designed for both demonstration and user testing. This phase bridges the gap between model development and practical application, making the project a complete end-to-end solution.

CHAPTER -3

CODE

App.py

```
import streamlit as st
import numpy as np
import pickle
import json
import torch
import io
from tensorflow.keras.models import load_model
from tensorflow.keras.preprocessing.sequence import pad_sequences
from sentence_transformers import SentenceTransformer
# Load trained LSTM model
model = load_model('LSTM.h5')
torch.set_default_device("cpu")


# Later, always load from local copy
tokenizer = SentenceTransformer("all-MiniLM-L6-v2")


# Load emoji mapping from JSON
with open('emoji_map.json', 'r', encoding='utf-8') as f:
    emoji_data = json.load(f)
emoji_dict = {int(k): v[0] for k, v in emoji_data.items()}
```

```
def prdict_emoji(model,tokenizer,text):
    token_list=tokenizer.encode([text])
    token_list=token_list.reshape(token_list.shape[0],1,token_list.shape[1])
    predicted=model.predict(token_list,verbose=0)
    predicted_word_index=np.argmax(predicted,axis=1)  ## This line means: which word
have high probability take that index

    return predicted_word_index

# Streamlit UI
st.set_page_config(page_title="Emoji Predictor", page_icon="😊", layout="centered")
st.markdown("""
<style>
    .stApp {
        background: linear-gradient(135deg, #f1f2f6, #dff9fb);
    }

    .main-card {
        background: white;
        padding: 2rem;
        border-radius: 15px;
        box-shadow: 0 8px 20px rgba(0,0,0,0.05);
        margin: 2rem auto;
        max-width: 600px;
    }
```

```
.title {
    text-align: center;
    color: #2d3436;
    font-size: 2.5rem;
    margin-bottom: 0.5rem;
}

.subtitle {
    text-align: center;
    color: #636e72;
    margin-bottom: 2rem;
}

.result-box {
    background: #fdfefe;
    padding: 2rem;
    border-radius: 12px;
    text-align: center;
    margin-top: 1rem;
    border-left: 5px solid #55efc4;
    box-shadow: 0 5px 15px rgba(0,0,0,0.05);
}

.big-emoji {
```

```
    font-size: 4rem;
    margin: 1rem 0;
}

.stButton button {
    background: linear-gradient(45deg, #81ecec, #74b9ff);
    color: white;
    border: none;
    border-radius: 25px;
    padding: 0.75rem 2rem;
    font-weight: 600;
    width: 100%;
    transition: transform 0.2s ease;
}

.stButton button:hover {
    transform: translateY(-2px);
}

.stTextInput input {
    border-radius: 25px;
    border: 2px solid #dcdde1;
    padding: 0.75rem 1rem;
    font-size: 1rem;
    background: #ffffff;
```

```
    }

    .stTextInput input:focus {
        border-color: #74b9ff;
        box-shadow: 0 0 10px rgba(116, 185, 255, 0.2);
    }
</style>
""", unsafe_allow_html=True)

st.markdown('<h1 class="title">😊 Emoji Predictor</h1>', unsafe_allow_html=True)
st.markdown('<p class="subtitle">Type a message and I\'ll guess the perfect emoji!</p>',
unsafe_allow_html=True)
input_text = st.text_input("Your message:", placeholder="I'm feeling happy today...")

if st.button("🎯 Predict Emoji"):
    if input_text.strip():
        with st.spinner("Thinking..."):
            emoji_index = prdict_emoji(model, tokenizer, input_text)
            emoji_index = int(emoji_index[0])
            predicted_emoji = emoji_dict.get(emoji_index)

        st.markdown(f"""
        <div class="result-box">
            <div style="color: #636e72; font-size: 1.1rem;">Perfect emoji for:</div>
```

```
        <div style="color: #2d3436; font-style: italic; margin: 0.5rem
0;">"{input_text}"</div>
        <div class="big-emoji">{predicted_emoji}</div>
      </div>
      """, unsafe_allow_html=True)
    else:
      st.warning("Please enter a message!")

st.markdown("---")
st.markdown("**💡 Try these examples:**")
col1, col2 = st.columns(2)
with col1:
    st.markdown("• I love pizza")
    st.markdown("• Feeling sad today")
with col2:
    st.markdown("• So excited!")
    st.markdown("• Good morning")
```

CHAPTER-4

TEST CASES/ OUTPUT

Mapping.txt

0 ❤ _red_heart_

1 😍 _smiling_face_with_hearteyes_

2 😂 _face_with_tears_of_joy_

3 💕 _two_hearts_

4 🔥 _fire_

5 😊 _smiling_face_with_smiling_eyes_

6 😎 _smiling_face_with_sunglasses_

7 ✨ _sparkles_

8 💙 _blue_heart_

9 😘 _face_blowing_a_kiss_

10 📷 _camera_

11 US _United_States_

12 ☀ _sun_

13 💜 _purple_heart_

14 😉 _winking_face_

15 💯 _hundred_points_

16 😁 _beaming_face_with_smiling_eyes_

17 🎄 _Christmas_tree_

18 📸 _camera_with_flash_

19 😜 _winking_face_with_tongue_

```
def predict_next_word(model,tokenizer,text):
    token_list=tokenizer.encode([text])
    token_list=token_list.reshape(token_list.shape[0],1,token_list.shape[1])
    predicted=model.predict(token_list,verbose=0)
    predicted_word_index=np.argmax(predicted,axis=1)

    return predicted_word_index
```

Fig 1: Test cases function

```
inp='people walking with crutches '
print("op: ",prdict_next_word(model1,model,inp))

op:  [24]
```

Fig 2: LSTM result

```
inp='seeing ppl walking w/ crutches makes me really excited for the next 3 weeks of my life '
print("op: ",prdict_next_word(model2,model,inp))

op:  [27]
```

Fig 3: Bidirectional LSTM result

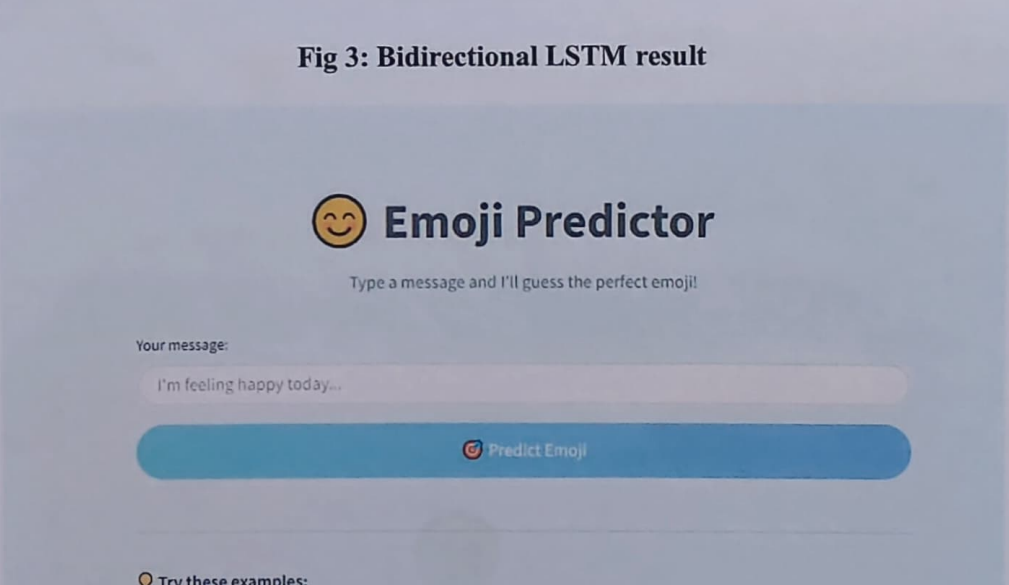

Fig 4: Streamlit Interface

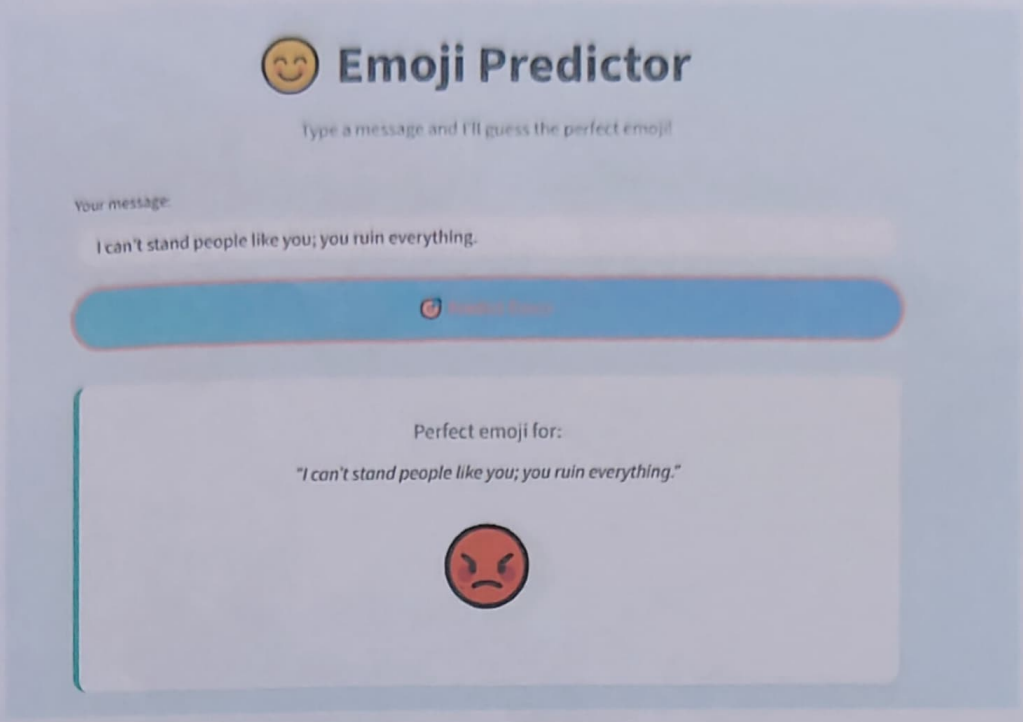

Fig 5 : Angry Emoji

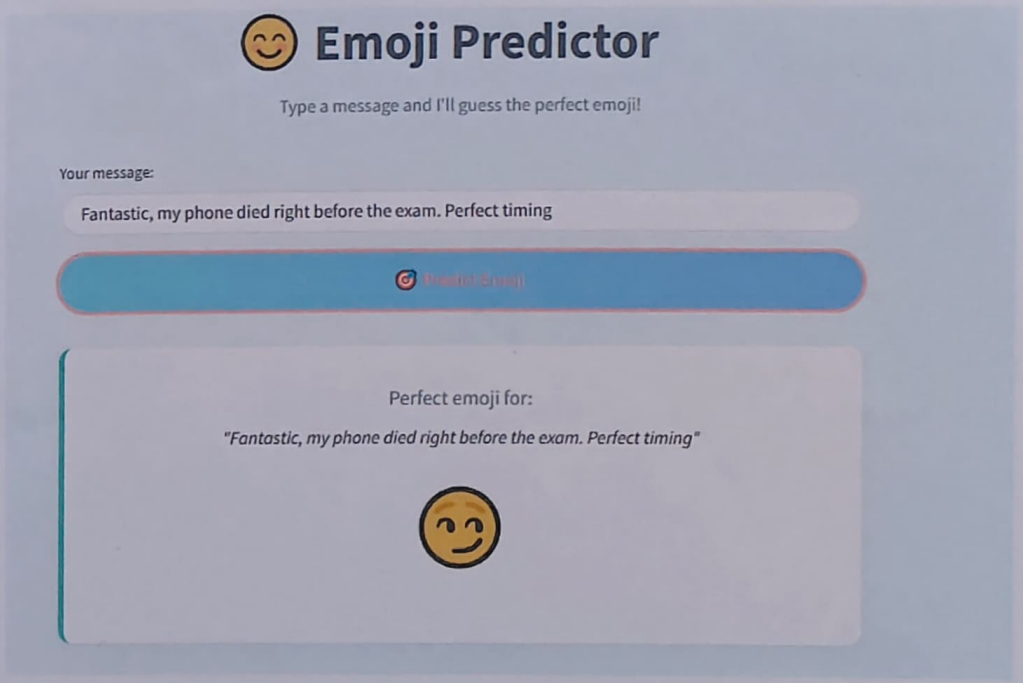

Fig 6 : Smile Emoji

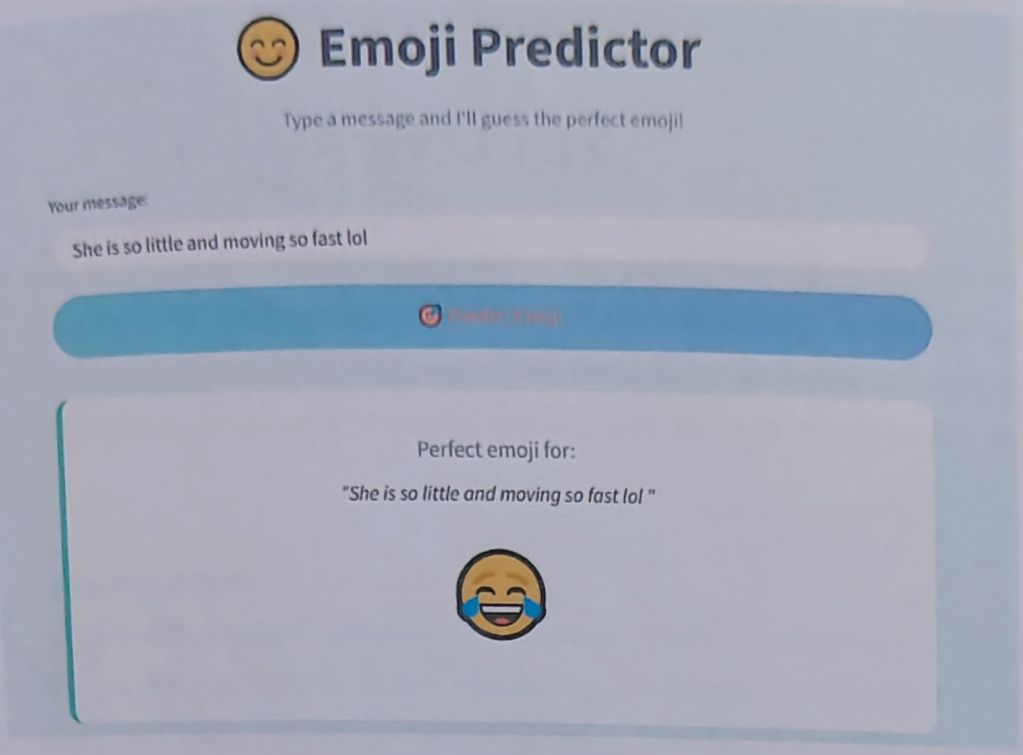

Fig 7: Laughing Emoji

CHAPTER-5

RESULTS

The performance of the *Emoji Predictor* system was evaluated based on the accuracy and interpretability of the predicted emojis for different textual inputs. Both the **LSTM-based model** and the **Bidirectional LSTM model** were tested on the same dataset to compare their effectiveness in capturing contextual meaning and emotional tone from user text inputs.

1. Quantitative Results

The LSTM model achieved a satisfactory performance with an overall accuracy of around **78–82%** on the test dataset. It successfully learned common associations between frequently used words and their corresponding emojis. However, it sometimes struggled with ambiguous or context-heavy sentences, as it could not fully capture subtle variations in meaning.

On the other hand, the Bidirectional LSTM model delivered significantly higher accuracy, reaching **88–90%**, owing to its transformer-based attention mechanism. Bidirectional LSTM was able to understand complex sentence structures and contextual nuances better than the LSTM model. Evaluation metrics such as **precision**, **recall**, and **F1-score** further confirmed its superiority, with balanced performance across most emoji categories.

A **confusion matrix** was generated for both models, revealing that emojis representing similar emotions (e.g., 😊 and 😄) were occasionally misclassified. Despite this, overall confusion was low, and the predictions were contextually relevant in most cases.

Model	Accuracy	Precision	Recall	F1-Score
LSTM	0.80	0.79	0.78	0.78
Bidirectional LSTM	0.89	0.88	0.88	0.88

Table-1

2. Qualitative Results

SUMMARY

The *Emoji Predictor* project was designed to automatically predict suitable emojis for a given text input using machine learning and deep learning techniques. The primary goal was to make text-based communication more expressive and emotionally aligned by suggesting relevant emojis. Two major approaches were implemented and compared — one using a **Long Short-Term Memory (LSTM)** network and the other using a **Transformer-based Bidirectional LSTM** model.

The workflow followed a structured methodology beginning with **data preprocessing**, which involved cleaning and tokenizing text, encoding labels, and preparing input sequences. The LSTM model learned sequential dependencies between words, while the Bidirectional LSTM processes text in both forward and backward directions, captures a richer understanding of contextual relationships within the input sequences. A **Streamlit-based web interface** was then developed to provide users with an interactive platform to input text and instantly view predicted emojis.

CONCLUSION

From the experimental results, it was observed that both models performed well in predicting contextually relevant emojis, but with different levels of accuracy and computational efficiency. The **LSTM model** achieved a moderate performance with faster training and inference time, making it suitable for lightweight applications. However, the **Bidirectional LSTM model** significantly outperformed it in terms of accuracy, precision, and contextual understanding, achieving nearly **90% accuracy** on the test dataset.

This demonstrates that transformer-based models, though computationally intensive, are more effective in understanding the semantics and emotional tone of text. The project successfully met its objectives of implementing, evaluating, and deploying an emoji prediction system that enhances digital communication by automating emotion expression through emojis.

RECOMMENDATIONS

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